Generating self-signed certificates

Snapdrive for Unix

Ivana Devine, Aksel Davis
January 21, 2021

This PDF was generated from https://docs.netapp.com/us-en/snapdrive-unix/linux-administration/task_generating_self_signed_certificates.html on August 06, 2021. Always check docs.netapp.com for the latest.

Table of Contents

Generating self-signed certificates 1

Generating self-signed certificates

The SnapDrive for UNIX daemon service requires that you generate a self-signed certificate for authentication. This authentication is required while communicating with the CLI.

Steps

1. Generate an RSA key:

   `$ openssl genrsa 1024 > host.key $ chmod 400 host.key`

   ```
   # openssl genrsa 1024 > host.key Generating
   RSA private key, 1024 bit long modulus
   ..............................++++++ ...++++++ e is 65537(0x10001)
   # chmod 400 host.key
   ```

2. Create the certificate:

   `$ openssl req -new -x509 -nodes -sha1 -days 365 -key host.key > host.cert`

   The `-new`, `-x509`, and `-nodes` options are used to create an unencrypted certificate. The `-days` option specifies the number of days the certificate remains valid.

3. When asked to fill out the certificate's x509 data, enter your local data:

   ```
   # openssl req -new -x509 -nodes -sha1 -days 365 -key host.key >
   host.cert
   You are about to be asked to enter information that will be incorporated
   into your certificate request.
   What you are about to enter is what is called a Distinguished Name or a
   DN. There are quite a few fields
   but you can leave some blank For some fields there will be a default
   value, If you enter '.', the field will be left blank.

   Country Name (2 letter code) [AU]:US
   State or Province Name (full name) [Some-State]:California
   Locality Name (eg, city) []:Sunnyvale
   Organization Name (eg, company) [Internet Widgits Pty Ltd]:abc.com
   Organizational Unit Name (eg, section) []:
   Common Name (eg, YOUR name) []:localhost
   Email Address []:postmaster@example.org
   ```

   The `Common Name` value must be *localhost*.

4. Extract metadata (optional).

```
$ openssl x509 -noout -fingerprint -text < host.cert > host.info
```

You can save the certificate metadata for quick reference later.

5. Combine key and certificate data.

   SnapDrive for UNIX requires the key and certificate data to be in the same file. The combined file must be protected as a key file.

   **$ cat host.cert host.key > host.pem **

   && rm host.key

   $ chmod 400 host.pem

```
# cat host.cert host.key >  /opt/NetApp/snapdrive.pem
# rm host.key rm: remove regular file `host.key'? y
# chmod 400 /opt/NetApp/snapdrive.pem
```

6. Add the complete path of the daemon certificate to the *`sdu-daemon-certificate-path`* variable of the `snapdrive.conf` file.

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.